Beyond Distance

A Scoping Review Uncovering Research Directions in Human–Robot Proxemics

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1 Supplementary Material

This document provides the PRISMA flowchart, the list of 25 papers identified for the scoping review, classified for each dimension, and the Sankey diagram.

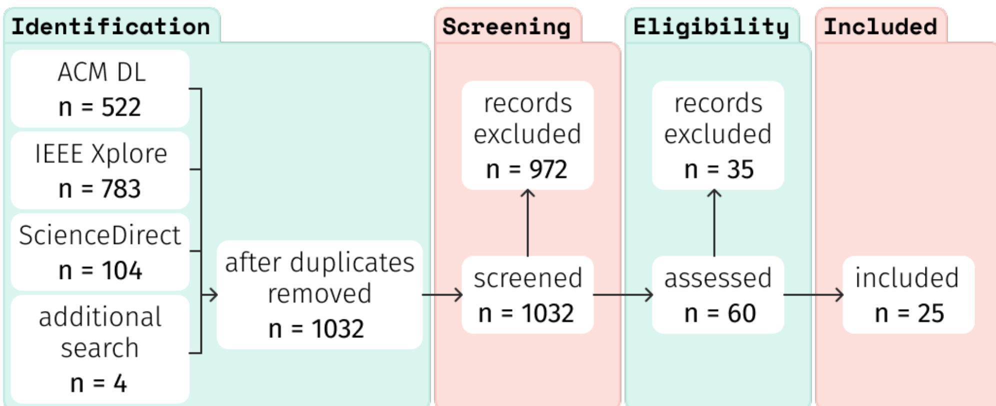

Fig. 1. PRISMA flowchart

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Table 1. Full corpus of papers.
Abbreviations used: *Robot-Rel Factors* = Robot-Related Factors Influencing Proxemic Responses, *Human-Rel Factors* = Human-Related Factors Influencing Proxemic Responses, *Computational Models* = Computational Models for Adaptive Behaviour, *Empirical Norms* = Empirical Proxemic Norms, *Soc & Env Influences* = Social and Environmental Influences

N.	Paper	Year	Venue	Participant Population	Interaction Context	Research Theme	Knowledge Contribution
1	Asavanant and Umemuro	2021	RO-MAN	Adults	Conversational	Proxemic Preferences	Robot-Rel Factors
2	Bhagya et al.	2019	RO-MAN	Adults	Conversational	Human Factors	Robot-Rel Factors
3	Camara et al.	2025	IROS	Adults	Navigation	Proxemic Preferences	Computational Models
4	Jost et al.	2021	ETFA	Adults	Collaborative	Human Factors	Human-Rel Factors
5	Kosiński et al.	2016	RO-MAN	Adults	Conversational	Proxemic Preferences	Computational Models
6	Lauckner et al.	2014	RO-MAN	Adults	Navigation	Proxemic Preferences	Empirical Norms
7	Lo et al.	2019	IROS	Adults	Navigation	Social Navigation	Navigation Strategies
8	Mizaridis et al.	2024	RO-MAN	Adults	Navigation	Non-Verbal Communication	Human-Rel Factors
9	Mumm and Mutlu	2011	HRI	Adults	Conversational	Proxemic Preferences	Robot-Rel Factors
10	Neggers et al.	2018	ICSR	Adults	Navigation	Proxemic Preferences	Empirical Norms
11	Obaid et al.	2016	RO-MAN	Adults	Conversational	Non-Verbal Communication	Robot-Rel Factors
12	Pacchierotti et al.	2006	RO-MAN	Adults	Navigation	Proxemic Preferences	Empirical Norms
13	Park et al.	2026	HRI	Adults	Navigation	Social Navigation	Empirical Norms
14	Rajamohan et al.	2019	RO-MAN	Adults	Conversational	Human Factors	Human-Rel Factors
15	Rossi et al.	2017	ICSR	Adults	Conversational	Proxemic Preferences	Human-Rel Factors
16	Ruijten and Cuijpers	2017	RO-MAN	Adults	Collaborative	Spatial Formations	Soc & Env Influences
17	Sorrentino et al.	2021	RO-MAN	Adults	Navigation	Non-Verbal Communication	Navigation Strategies
18	Taguchi et al.	2026	SII	Adults	Conversational	Human Factors	Robot-Rel Factors
19	Takayama and Pantofaru	2009	IROS	Adults	Conversational	Human Factors	Human-Rel Factors
20	Tokmurzina et al.	2018	HRI	Children	Educational	Child-Specific Proxemics	Human-Rel Factors
21	Tolksdorf et al.	2021	IDC	Children	Educational	Child-Specific Proxemics	Human-Rel Factors
22	Torta et al.	2013	Int. J. Soc. Robot.	Adults	Conversational	Proxemic Preferences	Computational Models
23	Trovato et al.	2018	RO-MAN	Adults	Navigation	Non-Verbal Communication	Robot-Rel Factors
24	Xu et al.	2025	ICRA	Adults	Navigation	Human Factors	Robot-Rel Factors
25	Yoo et al.	2026	HRI	Adults	Navigation	Social Navigation	Soc & Env Influences

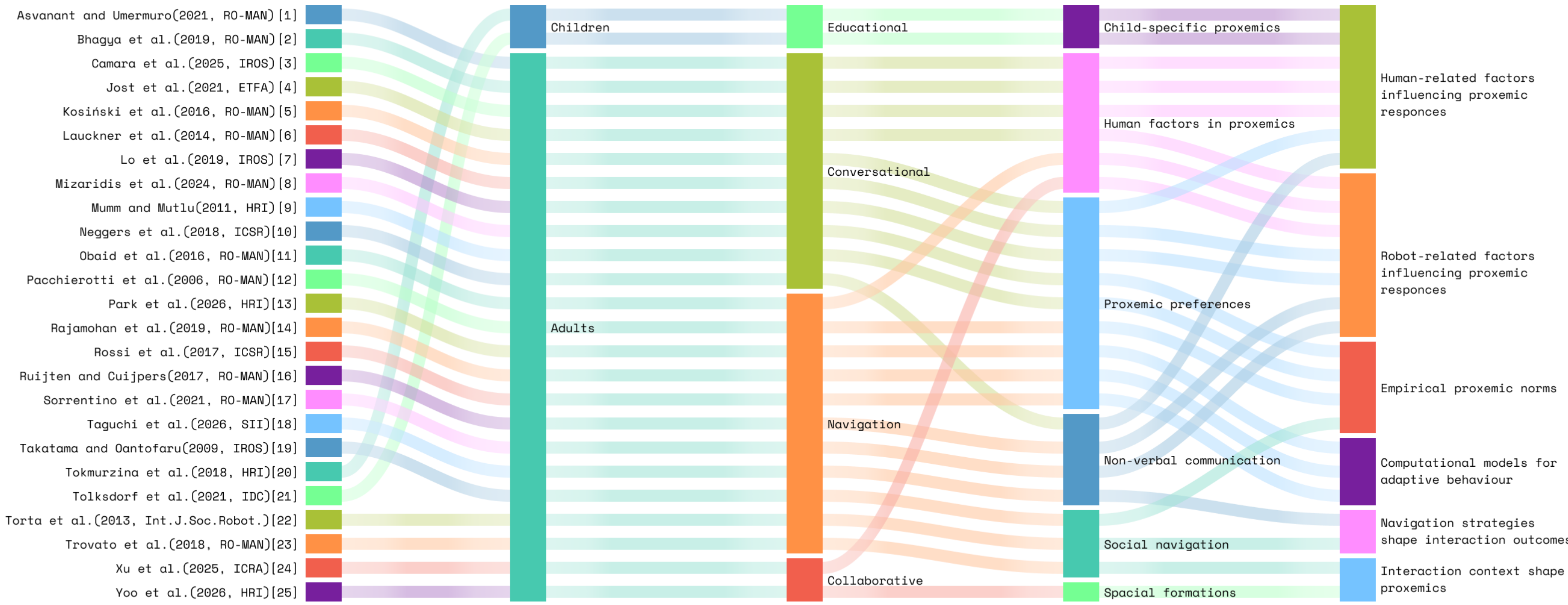

Fig. 2. Sankey diagram of the literature showing connections through *participant populations*, *interaction contexts*, *research themes* and *knowledge contributions* across the field.

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